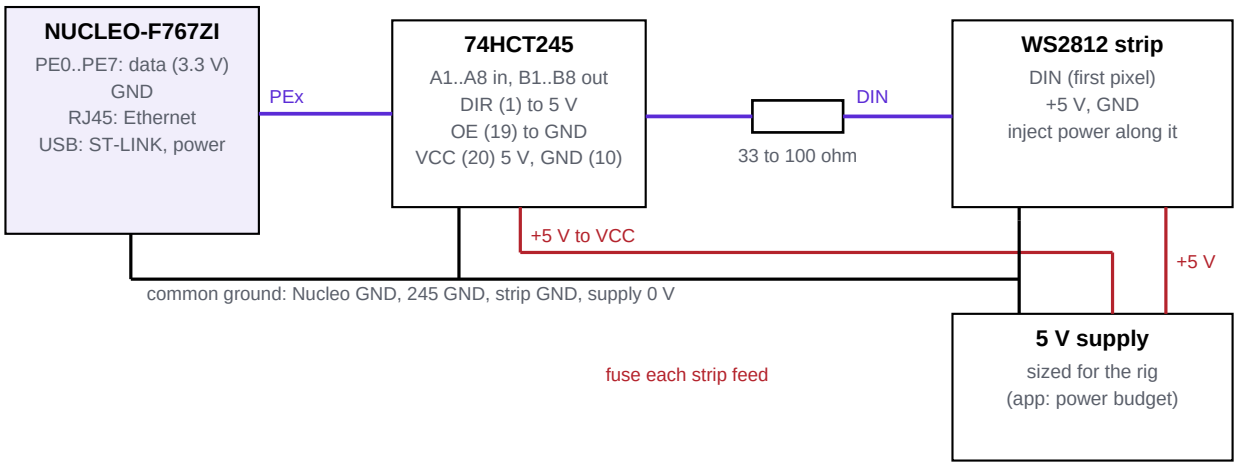
Psyight pixel node

Wiring, pinout, flashing and set-up for the NUCLEO-F767ZI running `firmware/node-f767` (stage F3, config protocol 3).

What	Where
Data lines (8)	PE0 to PE7, one WS2812-class strip chain each, through a 74HCT245
Network	RJ45 (CN14), 100 Mbit; Art-Net (UDP 6454) and sACN; config on UDP 6455
Programming and console	USB on the ST-LINK side (CN1): flashing, and a virtual COM port at 115200 8N1
Power for the board	USB (ST-LINK) for the bench; the strips never draw from the Nucleo

1. One data line, end to end



The MCU's outputs are 3.3 V; WS2812 inputs want at least 0.7 x VDD (3.5 V at 5 V), so each line goes through a 74HCT245 powered at 5 V (HCT inputs accept 3.3 V as high). A 33 to 100 ohm series resistor at the 245's output damps ringing on the cable. Keep the data cable short (a few metres) or mount the node near the strip; for long runs use RS-485 per line (not built yet).

2. Data line pins

Each line can be reached on a Zio connector (the Arduino-style headers) or on the ST morpho headers (the long double rows at the board's edges). Use one or the other for a pin, not both. The app's Preflight > Node setup numbers lines 1 to 8; line n is lane n-1 here.

Line (app)	Lane	STM32 pin	Zio connector	ST morpho	Zio name
1	0	PE0	CN10 pin 33	CN12 pin 64	D34
2	1	PE1	(not on Zio)	CN11 pin 61	-
3	2	PE2	CN9 pin 14	CN11 pin 46	D56
4	3	PE3	CN9 pin 22	CN11 pin 47	D60
5	4	PE4	CN9 pin 16	CN11 pin 48	D57
6	5	PE5	CN9 pin 18	CN11 pin 50	D58
7	6	PE6	CN9 pin 20	CN11 pin 62	D59
8	7	PE7	CN10 pin 20	CN12 pin 44	D41

- PE1 is only on the morpho header (CN11 pin 61).
- PE2 also appears on CN10 pin 25 (QSPI_BK1_IO2): CN9 pin 14 and CN10 pin 25 are the same pin. Wire only one of them.

- PE2 to PE6 double as the trace port (TRACECLK, TRACED0 to 3); nothing uses tracing here, so they are free.
- The default configuration drives lines 1 to 4 (the app's 2.5 by 1.2 m test frame, `active 15`); Node setup sets the lines the current rig needs.

3. Power, ground and the other pins the node uses

Signal	Zio	ST morpho	Notes
+5 V	CN8 pin 9	CN11 pin 18	Board 5 V. Do not feed strips from it.
+3.3 V	CN8 pin 7	CN11 pin 16	Not needed by the 245 (it runs at 5 V).
GND	CN8 pins 11, 13; CN9 pin 23; CN10 pins 5, 17, 22, 27	CN11 pins 8, 19, 20, 22, 49, 60; CN12 pins 9, 20, 39, 54, 63	Tie to the 245's GND and the strip supply's 0 V.
LD1 green	PB0		Blinks at 1 Hz while the firmware runs.
LD2 blue	PB7		On while show data arrives.
LD3 red	PB14		On without an Ethernet link, or after a fault.
Console	USART3: PD8 TX, PD9 RX		Wired to the ST-LINK: the USB virtual COM port, 115200 8N1.
Ethernet (RMII)	PA1, PA2, PA7, PB13, PC1, PC4, PC5, PG11, PG13		Onboard LAN8742A PHY. Jumpers JP6 and JP7 must be fitted (closed) for Ethernet.
Timer and DMA	TIM1, DMA2 (three streams into GPIOE BSRR)		Internal: no pins. TIM1's own pins (PE9, PE11, PE13) stay free.

4. Header maps (lane pins highlighted)

Pin 1 of each header is marked on the board's silkscreen; odd pins form one row and even pins the other. Only the stretch holding lane pins is drawn.

CN9 (Zio), pins 13 to 30

13	NC	**14**	**PE2 lane 2**
15	PA7	**16**	**PE4 lane 4**
17	PF2	**18**	**PE5 lane 5**
19	PF1	**20**	**PE6 lane 6**
21	PF0	**22**	**PE3 lane 3**
23	GND	24	PF8
25	PD0	26	PF7
27	PD1	28	PF9
29	PG0	30	PG1

CN10 (Zio), pins 16 to 33

17	GND	16	PG9
19	PD13	18	PE8
21	PD12	**20**	**PE7 lane 7**
23	PD11	22	GND
25	PE2 (= CN9 14)	24	PE10
27	GND	26	PE12
29	PA0	28	PE14
31	PB0	30	PE15
33	**PE0 lane 0**	32	PB10

CN11 (morpho), pins 43 to 62

43	PD6	44	PG3
45	PD7	**46**	**PE2 lane 2**
47	**PE3 lane 3**	**48**	**PE4 lane 4**
49	GND	**50**	**PE5 lane 5**
51	PF1	52	PF2
53	PF0	54	PF8
55	PD1	56	PF9
57	PD0	58	PG1
59	PG0	60	GND
61	**PE1 lane 1**	**62**	**PE6 lane 6**

CN12 (morpho), pins 39 to 64

39	GND	40	PE8
41	PD13	42	PF10
43	PD12	**44**	**PE7 lane 7**
45	PD11	46	PD14
47	PE10	48	PD15
49	PE12	50	PF14
51	PE14	52	PE9
53	PE15	54	GND
55	PE13	56	PE11
57	PF13	58	PF3
59	PF12	60	PF15
61	PG14	62	PF11
63	GND	**64**	**PE0 lane 0**

For the full sets the morpho headers are the easiest: lanes 1 to 6 sit together on CN11 (pins 46 to 62), lanes 0 and 7 on CN12 (pins 64 and 44), with grounds at CN11 pin 49 and CN12 pin 63 next to them.

5. 74HCT245 connections (DIP-20 or SO-20)

245 pin	Name	Connect to
1	DIR	+5 V (A to B)
2 to 9	A1 to A8	PE0 to PE7 (lane 0 on A1 ... lane 7 on A8)
10	GND	Common ground
18 down to 11	B1 to B8	Series resistor, then strip DIN (B1 for lane 0 ... B8 for lane 7)
19	OE (active low)	GND (always enabled)
20	VCC	+5 V, with a 100 nF capacitor to GND at the pin

6. Power and safety

- One ground for everything: the Nucleo, the 245 and every strip supply's 0 V.
- Fuse each output (each strip feed) for its cable and the strip's rating (SPEC 12.2, 13).
- Inject 5 V along long strips: a WS2812 draws up to about 60 mA at full white, so 150 pixels can draw 9 A. The app's power budget and brightness cap keep the show within the supply.
- Power the strips before (or with) the node, never data into an unpowered strip.
- The flash limiter in the app is on by default; it is a safeguard, not a certification.

7. Building and flashing the firmware

Needs STM32CubeIDE 1.19 (its arm-none-eabi-gcc 13.3 is found automatically), CMake 3.25 or later and Ninja. From `firmware/node-f767`:

```
powershell -ExecutionPolicy Bypass -File fetch_deps.ps1
cmake -S . -B build/f3 -G Ninja -DCMAKE_TOOLCHAIN_FILE=%CD%/cmake/arm-none-eabi.cmake
-DNODE_STAGE=3
cmake --build build/f3
```

Plug the Nucleo's ST-LINK USB (CN1) into the PC, then flash with STM32CubeProgrammer's command line (installed with STM32CubeIDE or on its own):

```
STM32_Programmer_CLI -l st-link
STM32_Programmer_CLI -c port=SWD -w build/f3/psyight_node_f3.elf -v -rst
```

- The first command lists the probe (it should say NUCLEO-F767ZI).
- The second erases only the sectors the program needs, writes, verifies and resets. The node's saved settings (the last flash sector) are kept; settings saved by protocol 2 firmware are read and upgraded at start.
- Without the command line: copy `psyight_node_f3.bin` onto the NODE_F767ZI USB drive the ST-LINK shows.
- Console: open the ST-LINK virtual COM port at 115200 8N1. Every 5 s it prints the address, output frames per second, packets, latches, sequence gaps, ignored universes, and "signal lost" while no show data arrives.

8. Setting the node up from the app

- Connect the node to the same network as the PC. It asks for an address by DHCP; without a lease it takes 2.x.y.z/8 (from its MAC) after 10 s. The console prints it.
- In Psyight: Preflight > Discover Art-Net nodes, then Node setup. It shows the data lines the current rig needs (pixels and colour order per line) and any notes, for example a line too long for 60 fps (more than 545 pixels).
- Choose what the node does when the signal stops (hold, fade, fallback show or blackout), after how long, and the fallback show's colours, brightness and speed. These are saved with the project.
- Set up the node: the app sends everything and applies it at once (no restart), then reads it back and lists any difference. Save on the node keeps it across restarts (output pauses 1 to 2 s while the flash sector erases).

Text protocol (UDP 6455), for scripts or a terminal: get, set <key> <value>, apply, save, defaults, reboot. Keys: universe, universes, pixels_per_universe, lane0 to lane7 (first pixel and count), order0 to order7 (rgb, grb, brg, rbg, gbr, bgr), active (line bit mask), loss_ms, policy, fade_ms, fallback_brightness and fallback_speed (thousandths), palette (one to four RRGGBB), name, long_name, protocols, ip, mask, gateway, dhcp.

9. Frame rate per line

Pixels per line	Universes	Frame time	Max frame rate
170	1	5.4 ms	about 185 fps
340	2	10.5 ms	about 95 fps
510	3	15.6 ms	about 64 fps
545	3.2	16.7 ms	60 fps (the show's rate)
680 (the most a line takes)	4	20.7 ms	about 48 fps

WS2812: 30 microseconds a pixel plus a 300 microsecond reset; all eight lines run in parallel, so the longest line sets the rate.

10. Bench checklist

- Scope PE0 with F1: bits 1.25 us (plus or minus 150 ns), 0 high 0.40 us, 1 high 0.80 us, frame gap at least 280 us; repeat with heavy network traffic.
- Link at 100 Mbit full duplex; unplug and replug: the node must come back.
- sACN multicast through the switch (IGMP snooping); every line changes on the same frame (sync universe 1000 or ArtSync).
- Stop sending: after loss_ms the chosen policy runs; check the fallback show's colours on each line (colour order per line, protocol 3).
- Set up from the app, apply without a restart (no glitch mid-frame), save, power cycle, read back.
- If the red LED stays on or the address stays 0.0.0.0: check the cable, then that JP6 and JP7 are fitted.

Sources: ST UM1974 Rev 7, Tables 11, 13 and 21 (NUCLEO-F767ZI); Psyight firmware sources (src/ws2812, src/board, src/net, src/core/node_config.h); SPEC sections 12 and 13. Generated by tools/python/make_node_guide.py.